



Bahir Dar Institute of Technology (BiT)

Faculty of Mechanical and Industrial Engineering

Laboratory Manual
Strength of Materials II

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Laboratory Report Writing

Effective teaching and learning of engineering involve seeing, handling, and manipulating real objects and materials. Engineering lab equipment allows students to interact directly with the data gathered and develop skills in a practical investigation. The students would get a first-hand learning experience by performing various experiments on their own. The presentation and report of any engineering or scientific work are probably as important as the actual work. Credit for discovery or development of new concepts cannot be received until the work has been described in a report or paper that is readily available for other people to read. Engineering laboratory reports must be written in a formal way from starting the experiment up to the final result. Therefore, the general structure of the report is:

1. Title
2. Objective
3. Theory
4. Apparatus
5. Procedure
6. Result
7. Calculations
8. Conclusion and discussion

1. Title

They are usually written:

- ✚ Name of the institute
- ✚ Name of the Faculty/school/ department
- ✚ Title of the experiment
- ✚ Name(s) of the students(s)
- ✚ Date of experiment
- ✚ No abbreviations.

2. Objectives

This is a brief statement of what you set out to determine, investigates, test or confirm.

3. Theory

For an engineering laboratory class, this will properly involve one or two explanations, and the statement of key mathematical expressions.

4. Apparatus

Alternative titles are equipment or experimental installations. In this section diagrams and words shall be imparted.

5. Procedure

This is the section in which you describe how you conducted the experiment. There is a well-established tradition of using the passive voice when describing an experimental procedure. In the passive voice, the object of a normal sentence (active voice) becomes the subject. So, describe your procedure using a ‘we did that’ or ‘I did that’ style. You will probably need to use the passive voice, but not for everything. Avoid the clumsier passive voice construction. Finally, keep your descriptions simple; don’t write more than is needed.

6. Result

If your experiment gave a reasonable number of results, it is probably best to present them in the form of a table. You must distinguish values that were actually measured and values that were calculated from the measurement. Remember to place a clear title at the top of each column defining the parameter, giving the symbol and units.

7. Calculation

Calculation of the relationship between parameters likely to involve plotting graphs, interpretation of graph, and perhaps comparisons with theory, will usually be the ‘aim’ of the experiment.

8. Discussion and Conclusion

The real purpose of laboratory classes is to allow you to learn something, so perhaps the conclusion should start ‘I learned the following ...’ but in your report, you have stated an aim (‘to determine...’, to investigate...,) and the conclusion should relate this. You should summarize the outcome of your experiment, what you have determined, or what the investigation has shown.

1. General Safety Guidelines

The following are a list of general safety rules based on the Occupational Safety and Health Administration (OSHA), to follow while working on the workshop. Although this is not a list of all requirements, following these rules may prevent you or a coworker from becoming involved in an incident.

- i. **Wear proper clothing for the type of work being done.** Do not wear loose-fitting clothes or jewelry that can get caught in moving parts. Do not roll up long sleeves. Do not wear highly flammable clothes. Do not wear sandals, open-toed or canvas shoes. Wearing safety toed shoes are preferable for protection of feet and toes.
- ii. **Remove** all jewelry, including rings, necklaces, bracelets, and watches that could get caught in tooling before starting work.
- iii. **Safety Glasses** must be worn at all times in the shop. Some operations may require additional eye protection or other personal protective equipment
- iv. **Hair.** Pull back long hair in a band and tuck it under your shirt collar or a cap to keep it from getting caught in tools. Long beards must also be contained.
- v. **Keep the work area clean.** Keep the floor free of scraps and oil. Cluttered work areas invite accidents. Keeping workshop and storage spaces clean and dry can help prevent many accidents. Sparks can ignite scraps, sawdust, and solvents. Water can conduct electricity. Do not stand in water, on damp floors or in the rain when working with electrical tools. Keep hands and tools dry.
- vi. The use of **food and drinks** are forbidden in the laboratory.
- vii. **Cell phones** and other personal electronic devices must not be used with working at any apparatus or machine. Loud music is prohibited.
- viii. **Use the correct tools for the job.** Do not use a tool or attachment for something it was not designed to do. Select the correct bit, cutter or grinding wheel for the material with which you are working. This saves time and improves the quality of work and reduces the risk of mishap. If necessary, consult the instructions or shop manual, or call a dealer or an expert on tool use.
- ix. **Know the machine.** Before using any tool, read the operator's manual, or comparable literature as available, to learn the applications, limitations, and potential of each power

tool. Never use a tool unless trained to do so. Inspect it before each use and replace or repair if parts are worn or damaged. Repair tools only if you are trained to do so. Inspect screws, nuts, bolts, and movable parts to make sure they are tightened. Make sure the cord will not become caught or tangled. The cord should be flexible, but not easy to knot. Clean the cord regularly and inspect the grounding connections. Use a ground fault circuit interrupter when working with power tools.

- x. **Know the switch location(s).** Remember where the switch is located so you can turn off the machine quickly.
- xi. **Fires.** Shop fires can be any, or all, of three fire classes: Class A, ordinary combustibles; Class B, combustible liquids; and Class C, fires in live electrical equipment. To eliminate the need for extinguishers for all three classes, install at least one 20-pound ABC Class extinguisher in a convenient location, and possibly more, depending on the size of the shop.
- xii. **Storage.** Develop a system of racks, bins and tool panels to make it easy to find the right tool or materials quickly. Don't store tools, supplies or spare parts in the aisle or on the floor where they become tripping hazards. Keep other flammable materials away from heaters and welding areas to prevent fire. Grease, oil, paint, and solvents should be stored in a closed metal container, preferably in metal cabinets. Gasoline or other fuels should never be stored inside the shop. Supplies and equipment should be stored in an area designed specifically for them.
- xiii. **Do not use** damaged equipment or equipment that does not appear to be operating normally.
- xiv. **Immediately report** any problems or concerns.

3. Unsymmetrical Cantilever Unit Manual

3.1. Introduction

Unsymmetrical bending is the structural loading in which the bending moment has components in the two principal axes of inertia. The principal axes of the cross-section of the beam do not coincide with the direction of the load, so the beam experiences combined bending and torsion stresses.

This type of bending is the one supported by cantilever beams, characterized by being fixed only on one of its ends. The main application of these beams is the building of bridges, pillars, etc.

The unsymmetrical cantilever unit (MVV) is designed to demonstrate the unsymmetrical bending in beams with easy tests that determine the deflection in the free end of the beam.

3.2. Objective

The objective of the apparatuses unit to determine the deflection of the beam according to the moment applied at its free end and for that purpose, we will change the moment applied changing the intensity of the load and/or its direction. The asymmetry of the beam will give different flexion for different directions of force application. The other objective is to determine the shear center of a cantilever beam and the position of the center of torsion of a cantilever beam

3.2. Background

3.1.1. Theory

Bending is the strain of some structural elements subjected to a bending moment in a direction perpendicular to its longitudinal axis and caused by the application of loads and by its own weight.

There are two types of bending: symmetrical and unsymmetrical. Symmetrical bending is that bending in which the bending moment acts only in one of the principal axes of the beam, being M_y or M_z .

Nevertheless, in the unsymmetrical bending, the bending moment has a component in both principal axes of the beam, M_y and M_z , thus combining torsion and bending stresses. Symmetrical bending

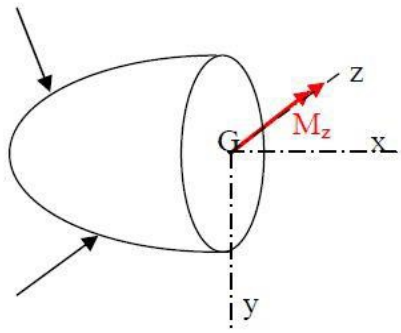


Figure 1: Symmetrical bending

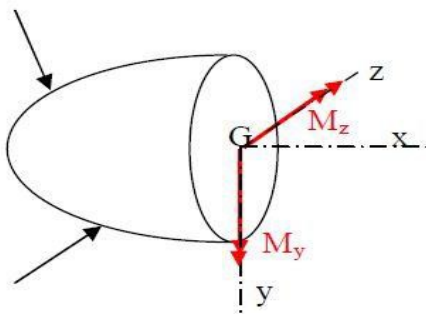


Figure 2: Unsymmetrical bending

3.1.2 Calculation of the unsymmetrical bending

As it has already been mentioned, the unsymmetrical bending is that bending where there are bending stresses in both principal directions of a beam.

To solve cases where unsymmetrical bending appears, it will be necessary to describe and solve the bending moments M_y and M_z , being these two particular cases of symmetrical bending. Afterward, the principle of superposition or vector addition of both moments will be applied thanks to the particularization of Navier's equation, which is defined by:

$$\sigma_x = -\frac{M_z(x)}{I_z} y + \frac{M_y(x)}{I_y} z$$

The principal axes of the beam are those axes where maximum and minimum moments of inertia take place. The moments of inertia are related to the maximum tensions and strains generated by

the bending stresses in the beam. Therefore, this value determines the maximum resistance of a structural element under bending together with the properties of that material.

To calculate the moments of inertia and strains and tensions, the graphical solution method known as Mohr's circle is normally used:

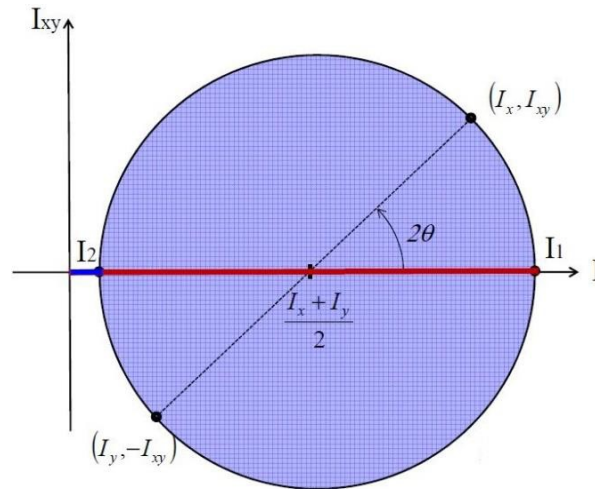


Figure 3: determination of I by Mohr's circle

The equations required to determine the moments of inertia are deduced from the graph:

$$I_1 = \frac{I_x + I_y}{2} + \sqrt{\left(\frac{I_x - I_y}{2}\right)^2 + I_{xy}^2}$$

$$I_2 = \frac{I_x + I_y}{2} - \sqrt{\left(\frac{I_x - I_y}{2}\right)^2 + I_{xy}^2}$$

$$\operatorname{tg}(2\theta) = -\frac{2I_{xy}}{I_x - I_y}$$

Translating the results of the moments of inertia obtained to the formula of Nervier particularization, the stress of the body in relation to its bending moments will be obtained:

$$\sigma_x = -\frac{M_1(x)}{I_1} \text{ and } +\frac{M_2(x)}{I_2} z$$

$$\text{Being: } M_z = M_1 = M \cdot \cos \theta$$

$$M_v = M_2 = M \cdot \sin \theta$$

To find the horizontal and vertical deflections generated in the beam by the application of loads, according to their principal axes, the equation of the elastic curve must be solved. It is the geometrical place of the gravity center of the section after the deformation. The calculation of the beam deflections is defined by:

$$\Delta_x = \frac{M_1}{EI_1} \cdot \frac{W \cdot l^3 \cdot \cos \theta}{3 \cdot E \cdot I_1} \quad ; \quad \Delta_y = \frac{M_2}{EI_2} \cdot \frac{W \cdot l^3 \cdot \sin \theta}{3 \cdot E \cdot I_2}$$

The result of the deflections for the principal axes of the beam is obtained with it. Translating this to the real axes, the deflections are defined according to:

$$\Delta_U = \Delta_x \cos \theta + \Delta_y \sin \theta$$

$$\Delta_V = \Delta_y \cos \theta - \Delta_x \sin \theta$$

3.1.3 Shear center

The center of torsion of the beam is a point O of the transverse section of the beam through which the resultant of the shear stress due to simple bending passes. Any shear stress that passes through the center of torsion will not generate a moment of torsion in the cross-section. But every shear stress generates a moment of torsion given by the distance of the shear stress to the center of torsion. The geometric position of the center of torsion is given by (y_T and z_T). When there is a symmetry axis, the center of torsion is located on it. In pieces with two symmetry axes the center of torsion coincides with the center of gravity of the section and in that case, the bending and the torsion are not coupled, the beam can have bending but not torsion and vice versa. In beams with asymmetries in their cross-section, it is necessary to determine the center of torsion in order to determine the stresses correctly.

When using the x coordinate to measure distances along the axis of a Prismatic piece and the coordinates (y, z) for the coordinate of any point on a cross-section, the center of torsion is the point defined by the coordinates (y_T, z_T) given by:

$$y_T = \frac{I_z \cdot I_{y\bar{w}} - I_{yz} \cdot I_{z\bar{w}}}{I_z \cdot I_y - I_{yz}^2} \quad z_T = \frac{I_y \cdot I_{z\bar{w}} - I_{yz} \cdot I_{y\bar{w}}}{I_z \cdot I_y - I_{yz}^2}$$

Where I_y and I_z and I_{yz} are the moments of area and the product of inertia. And where $I_{y\bar{w}}$ and $I_{z\bar{w}}$ are the products of sectorial inertia defined as:

$$I_{y\bar{w}} = \int_A z w_0(y, z) dy dz$$

$$I_{z\bar{w}} = \int_A y w_0(y, z) dy dz$$

And $W_o(y, z)$ is the auxiliary function of a unit warp.

It is important to indicate that:

- If the axis Y is a symmetry axis of the cross-section, then, Z_T and Z_G .
- If the axis Z is a symmetry axis of the cross-section, then Y_T and Y_G .
- If a piece has two axes of symmetry, Y and Z, and they are considered
- Barycentric coordinates, then $Y_T = Y_G = 0$ and $Z_T = Z_G = 0$.

3.2. Description of the unit

3.2.1 Unsymmetrical bending unit:

14 15

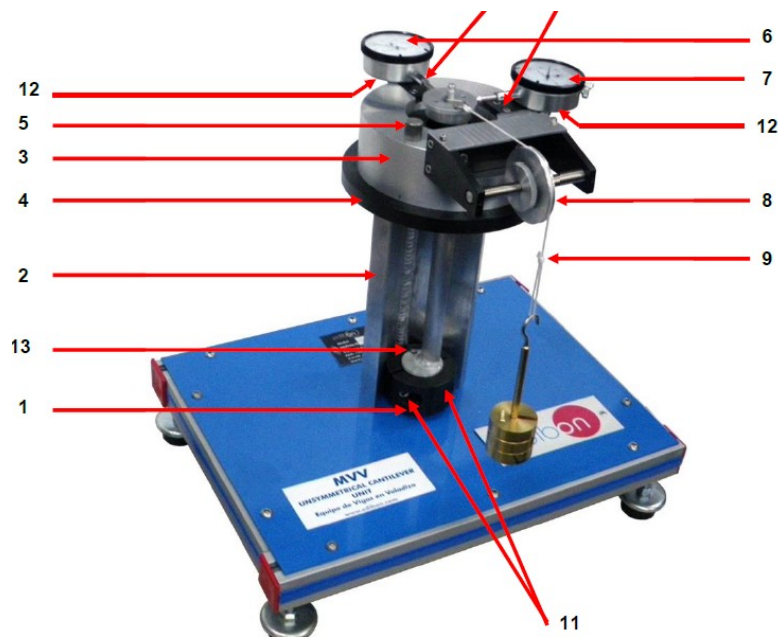
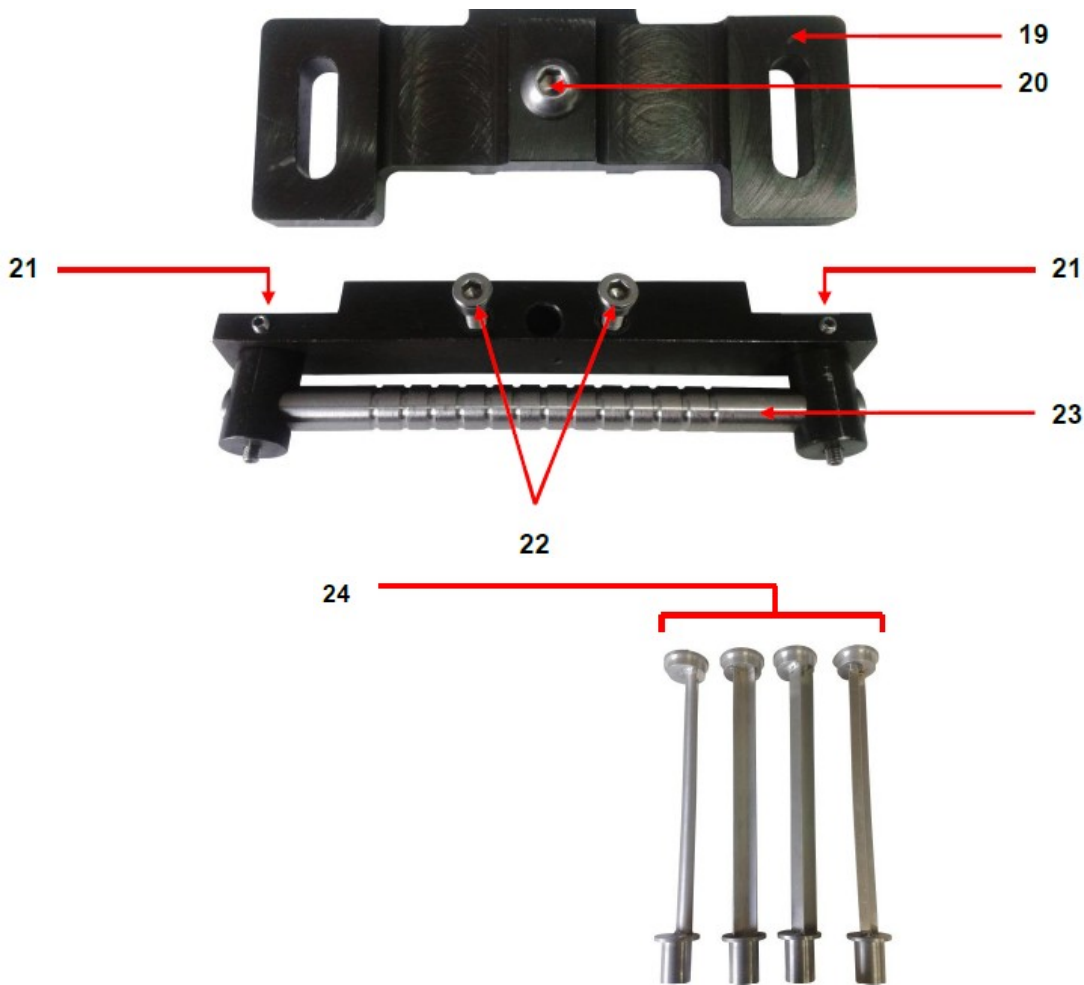


Figure 4: Unsymmetrical bending unit:



1. A-1: Clamp for beams anchorage.
2. C-1: Main column of the unit.
3. C-2: Loading head of the unit.
4. D-1: Disc with notches, from 0° to 180° .
5. P-1: Pin to fix C-2 at the chosen angle in D-1.
6. R-1: Dial gauge to measure vertical deflections, ΔU .
7. R-2: Dial gauge to measure horizontal deflections, ΔV .
8. PO-1: Pulley, located in C-2, to apply loads.
9. CU-1: Cord to hang the loads, fastened to the beam, V-1 or V-2, which passes through PO-1.
10. V-1: Beams made of steel.
11. T-1: Screws to fix V-1 or V-2 to A-1.
12. T-2: Screws to position R-1 and R-2.
13. CL-1: Peg to avoid movements of the beams during the tests.
14. S-1: Support for R-1.
15. S-2: Support for R-2.
16. Allen key *no. 6* to fasten the screws (T-1) of the clamp (A-1) to fix the beams.
17. Allen key *no. 8* to fasten and loosen the pin (P-1).
18. Fixed wrench *no. 12-13* to fasten the screws (T-2) to adjust the dial gauges (R-1, R-2).



- 19. S-3: Support of the accessory to locate R-1 and R-2.
- 20. T-3: Screw to fasten S-3 to the unit, fixing it to S-1.
- 21. D-2: Flat discs where the dial gauges are supported.
- 22. T-4: Screws to fasten the accessory to the test beam.
- 23. B-1: Bar with notches to position CU-1 for the torsion center test.
- 24. V-2: Test beams made of aluminum.

3.2.1. Description of the mechanism

The unsymmetrical cantilever beam, MVV, allows demonstrating the unsymmetrical bending in beams through easy tests that determine the deflections in the free end of the beam. It consists of a vertical cantilever, fixed by a clamp, A-1, at its lower end to the main column, C-1, which has a heavy base. It has a loading head, C-2, located at the top of the main column, which can freely

rotate at intervals of 15° about the vertical axis of the beam. It facilitates the study of deflections in different directions.

The tests are based on the application of a horizontal load in the free end of the beam with a cord, CU-1, attached to the beam. The cord passes over a pulley, PO-1, located in the loading head. Two dial gauges, R-1 and R-2, measure the deflections ΔU and ΔV , respectively, supported by the beam when loads are located.

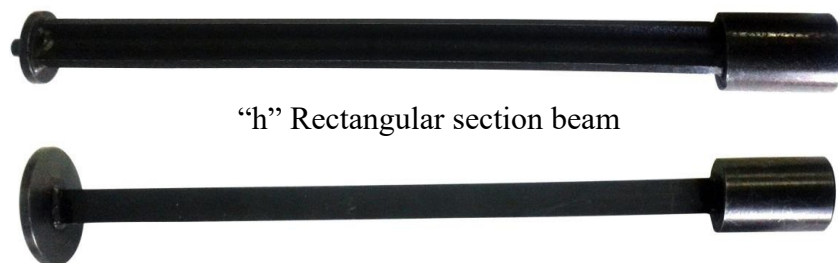
3.2.2. Practical possibilities

- To determine the deflections of the beam according to the moment applied at its free end.
 - ✓ Modifying the direction of the load application.
 - ✓ Modifying the intensity of the applied load.
- To determine the center of the torsion of a cantilever (with the accessory MVV/1).

3.2.3. Unsymmetrical bending unit (MVV) specifications

The MVV unit consists mainly of:

- Main column and loading head of the units.
- Clamp to fix the stainless steel beams.
- Disc with slots from 0° to 180° , divided into intervals of 15° , to situate the load according to the angle chosen for the test.
- Pulley for the load application.
- Two flat contact point dial gauges of 25mm travel and 0.01mm precision.
- Two test beams made of steel.
 - ✓ “L” section beam.



- Allen key *no. 6* to fasten the screws of the clamp to fix the beams.
- Allen key *no. 8* to fasten and loosen the pin.
- Fixed wrench *no. 12-13* to fasten the screws to adjust the dial gauges.

3.2.4 MVV/1 specifications

The MVV/1 unit consists of:

- Support of the accessory for the dial gauges.
- Flat discs of the accessory where the dial gauges employed to measure deflections are supported.
- Bar with slots in different positions to perform tests for the study of the center of torsion.
- Pulley for the load application at different angles.
- Four beams made of aluminum:

“C” section beam



“L” section beam



“U” section beam.



“Zigzag” section beam.



3.2.5 Dimensions and weight

Height: 400 mm, Width: 300 mm, Length: 400 mm, Weight: 14 kg.

3.2.6 Required services (not included in the unit)

2 sets of weights “type B” (Set B).

Each “type B” set consists of:

- 6 weights of 200 grams
- 6 weights of 100 gram
- 2 weights of 50 gram
- 2 weights of 20 gram
- 2 weights of 10gram
- 1 hook of 100 gram

3.3 Experimental Procedure and Practical Exercises

3.3.1. Laboratory Practical Exercises

3.3.2 Practical exercise 1:

Determination of the deflection of the beam according to the moment applied at its free end

3.3.3Objective

The objective of this practical exercise is to determine the deflections of the beam according to the moment applied on its free end. For that purpose, we will change the moment applied to change the intensity of the load and/or its direction. The asymmetry of the beam will give different flexion for different directions of force application.

3.3.4Required elements

- MVV unit
- “Type B” set of weights (set B)

3.3.5Experimental procedure

1. Remove the dial gauges, R-1 and R-2, from the unit before starting.
2. Rotate the loading head, C-2, in front of the slot of 0° of the disc, D-1, and introduce the pin, P-1, in D-1. Thread it until the load head is fastened and there is no play.
3. Select the beam to be tested, V-1.
4. Locate the beam in the clamp, A-1, and fix it with the help of the screws, T-1, in such a way that the slot of the beam at the lower end is blocked with the peg, CL-1, of the clamp.
5. Locate the dial gauges, R-1 and R-2, in the supports, S-1 and S-2, being one dial gauge perpendicular with respect to the other. Adjust them until their contact points touch the beam and the dial gauge indicates approximately half of its travel, 5mm.

6. Position the pulley, PO-1, in the center of its support.
7. Locate the cord, CU-1, over the pulley and tie it to the projection of the beam at its upper end.
8. Adjust the dial gauges, R-1 and R-2, rotating their upper scale until reaching the position 0.00 mm.
9. Hang and remove a load from the cord to check that the dial gauge returns to approximately 0mm. Otherwise, knock the beam slightly to stabilize it at its rest position.
10. Hang 0.5 kg on the cord, CU-1, and record the data obtained from the dial gauges, AV and AU, in the attached table.
11. Unscrew the pin, P-1, and remove it to rotate the disc, D-1. Fix the disc at the next slot and fasten it again with the pin, P-1.
12. Repeat the steps from 8 to 12 until reaching the position 180° of the disc, D-1, completing all possible angles.
13. Represent the deflections recorded in step 10 in an angle-deflection graph.
14. Repeat the steps from 8 to 14 with loads of 1 kg and 1.5 kg.
15. Repeat every step described with every beam supplied with the unit

3.3.6 Working sheet

Student: _____ Year: _____

Practical exercise: _____

1. Record the results obtained:

Type of beam:

	0.5 kg		1.0 kg		1.5 kg	
ANGLE (°)	Δu (mm)	Δv (mm)	Δu (mm)	Δv (mm)	Δu (mm)	Δv (mm)
0						
15						
30						
45						
60						
75						
90						
105						
120						
135						
150						
165						
180						

2. Plot the angle-deflection graph for each load:

Graph for a load of _____ kg

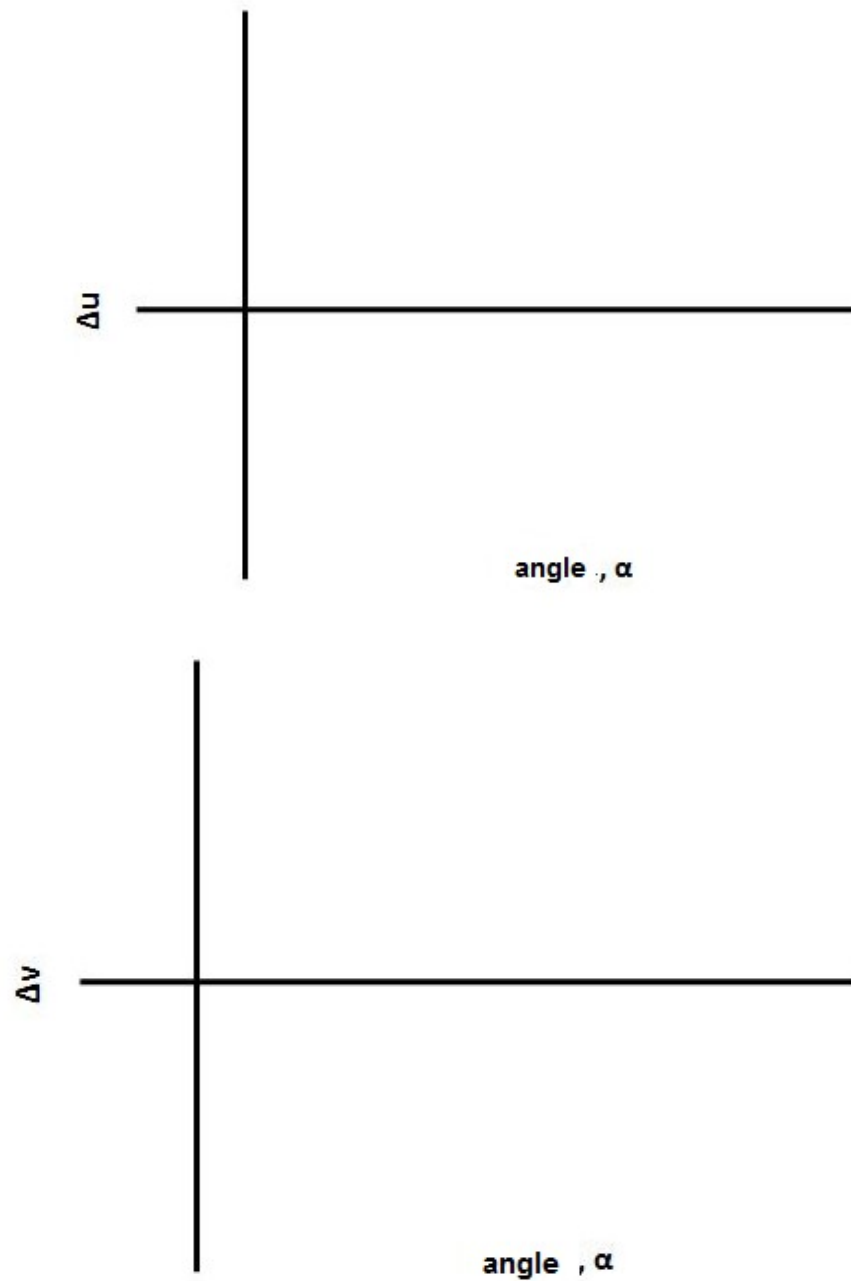


Figure 5: angle-deflection graph

3.4. Practical exercise 2:

Determination of the shear center of a cantilever beam

3.4.1. Objective

The objective of this practical exercise is to determine the position of the center of torsion of a cantilever, checking that the forces applied on it do not generate any rotation in the beam.

3.4.2. Required elements

- ✓ MVV unit
- ✓ MVV/1 accessory
- ✓ “Type B” set of weights (set B)

3.4.3. Experimental procedure

1. Remove the dial gauges, R-1 and R-2, from the unit before starting.
2. Put the loading head, C-2, in the slot of 90. of the disc, D-1.
3. Introduce the pin, P-1, in D-1 threading it to fix the loading head until there is no play.
4. Select the beam to be tested, V-1.
5. Put the support, S-3, of the MVV/1 accessory in the support, S-1, of the unit and fasten it at its lower side with the screw T-3.
6. Locate the beam in the clamp, A-1, and fix it with the help of the screws, T-1, in such a way that the two threads the beam has at its upper side are

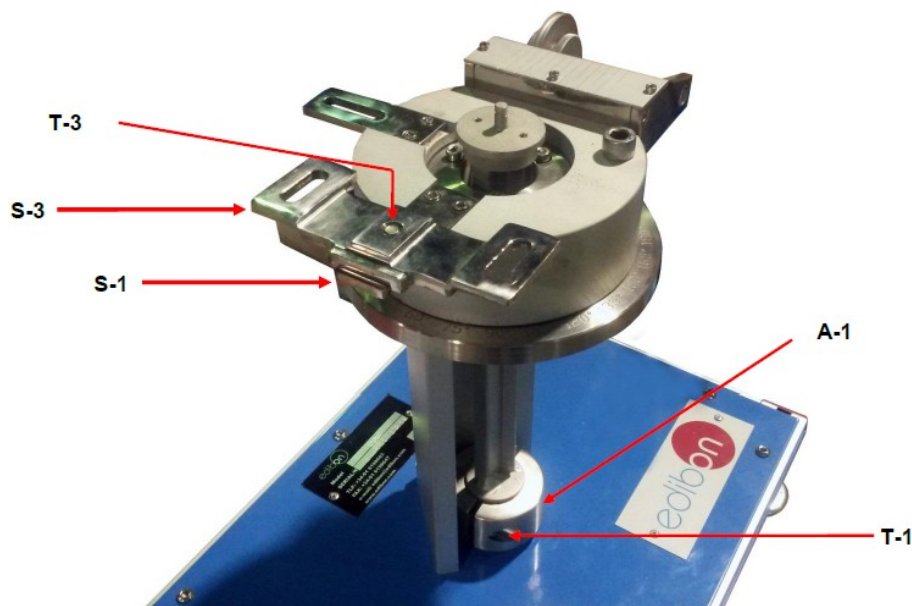


Figure 6: unit body parts

7. Fix the slotted bar, B-1, on the beam, V-1, and fasten it with the screws, T-4.
8. Locate the dial gauges in the support, S-3, with the screws T-2 in such a way that the contact point of the dial gauges, R-1 and R-2, lay on the flat discs, D-2, and they indicate approximately half of their travel, 5mm.



Figure 7: unit body parts

9. Locate the cord, CU-1, on the bar of the accessory, B-1, in the slot located further to the left for the test.
10. Adjust the pulley, PO-1, to the height of the cord, CU-1, and pass it over the pulley.
11. Adjust the dial gauges, R-1 and R-2, rotating their upper scale until reaching the position 0.00 mm.
12. Hang and remove a load from the cord to check that the dial gauge returns to approximately 0mm. Otherwise, knock the beam slightly to stabilize it at its rest position.
13. Hang 0.5 kg on the cord, CU-1, and record the data obtained, AU, in the attached table.
14. Remove the load from the cord, CU-1.
15. Put the cord, CU-1, in the following slot to the right of the accessory bar, B-1.
16. Locate the pulley, PO-1, in such a way that it is aligned with the slot where the cord was located.
17. Repeat the steps from 9 to 16 until reaching the position of the slot further to the right of the bar B-1.
16. Represent the deflections recorded in step 13 in an angle-deflection graph.
17. Repeat the steps from 9 to 18 with loads of 1 kg and 1.5 kg.
18. Repeat every step described with every beam made of aluminum, V-2, supplied with the unit

3.4.4. Working sheet

Student: _____ Year: _____

Practical exercise: _____

Record the results obtained:**Type of beam:**

	0.5 kg		1.0 kg		1.5 kg	
ANGLE (°)	Δu (mm)	Δv (mm)	Δu (mm)	Δv (mm)	Δu (mm)	Δv (mm)
0						
15						
30						
45						
60						
75						
90						
105						
120						
135						
150						
165						
180						

Plot the angle-deflection graph for each load:

Graph for a load of----- kg

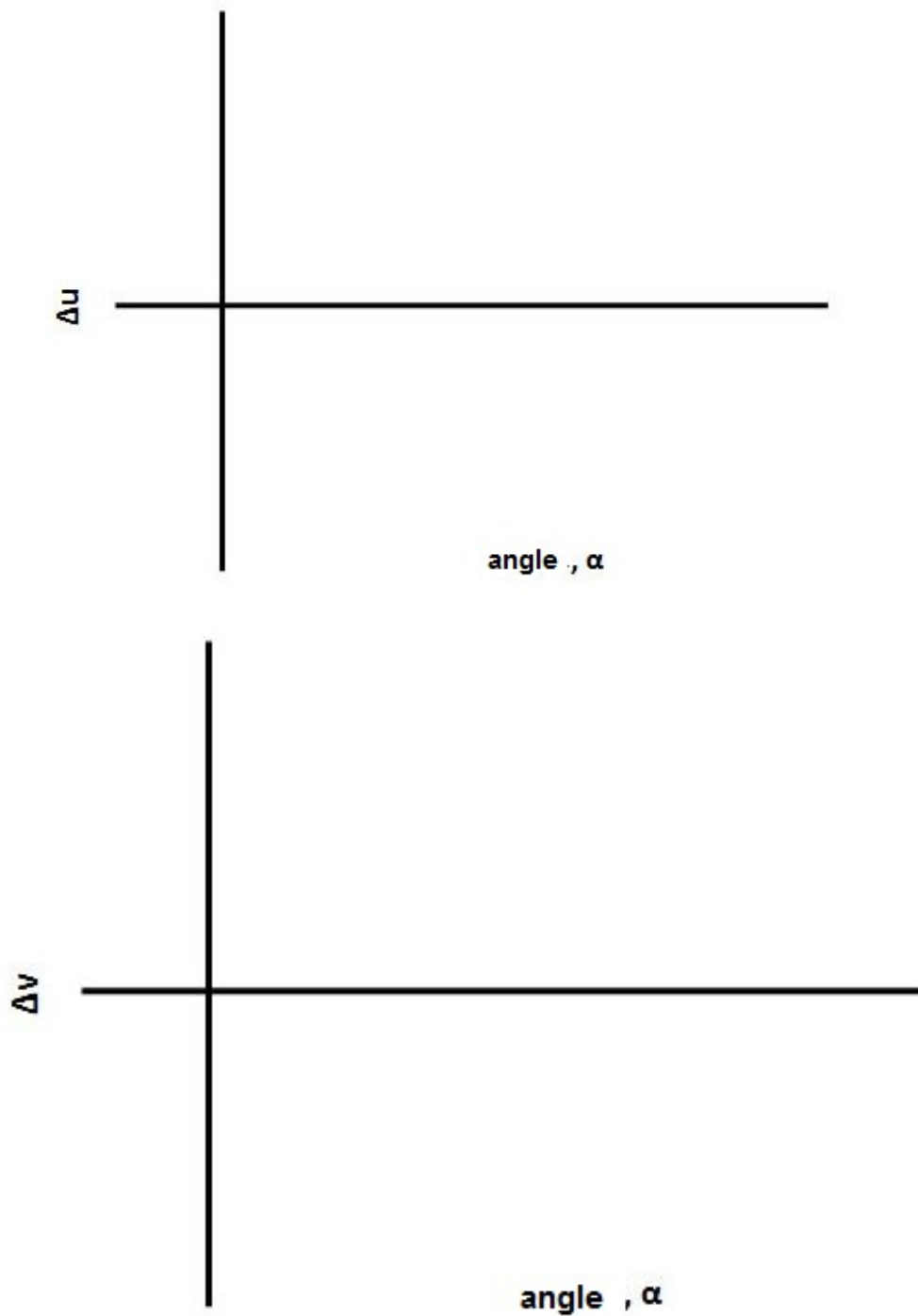


Figure 8: angle-deflection graph

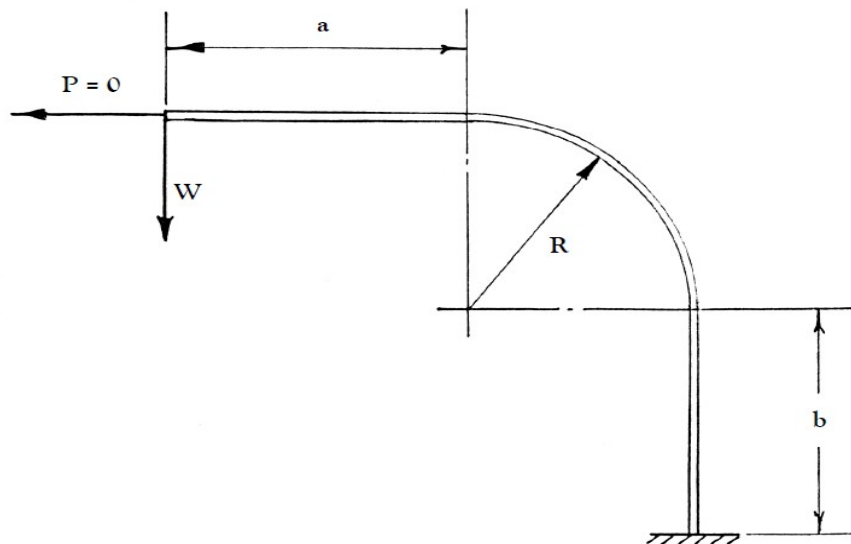
4. Curved beam deflection

4.1. Objects

1 To determine the experimentally the horizontally and vertical displacements at the free end of various thin curved bars when subject to a single concentrated load.

4.2. Background

Theory Revision Notes



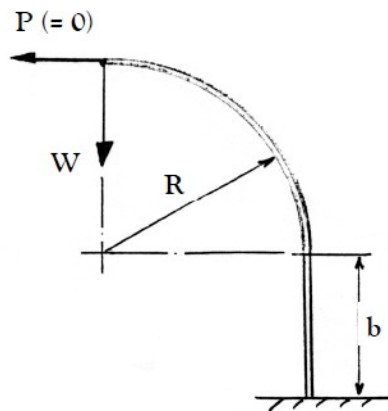
Assuming uniform section, then it can be shown that,

$$\Delta_W = \frac{Wa^3}{3EI} + \frac{WR}{EI} \left[\frac{\pi a^3}{2} + \frac{\pi R^2}{4} + 2aR \right] + \frac{W}{EI} \left[a^2b + 2abR + bR^2 \right]$$

&

$$\Delta_P = \frac{WR^2}{EI} \left[a(\pi/2 - 1) + \frac{R}{2} \right] + \frac{W}{EI} \left[abR + bR^2 + \frac{ab^2}{2} + \frac{b^2R}{2} \right]$$

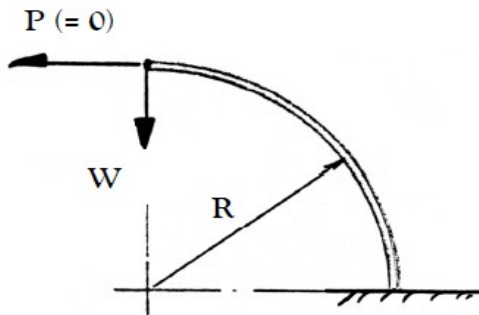
(i)

When $a = 0$

$$\Delta_W = \frac{\pi WR^3}{4EI} + \frac{WbR^2}{EI}$$

$$\Delta_P = \frac{WR^3}{2EI} + \frac{WbR}{EI} (R + b/2)$$

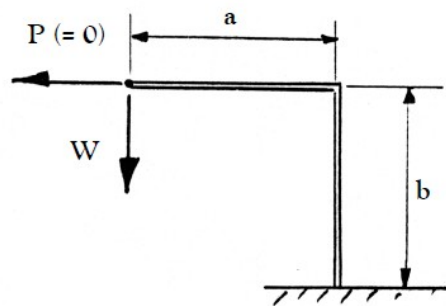
(ii)

When $a = 0$ & $b = 0$

$$\Delta_W = \frac{\pi WR^3}{4EI}$$

$$\Delta_P = \frac{WR^3}{2EI}$$

(iii)

When $R = 0$

$$\Delta_W = \frac{Wa^2}{EI} (a/3 + b)$$

$$\Delta_P = \frac{Wab^2}{2EI}$$

4.3. Tools and Equipment

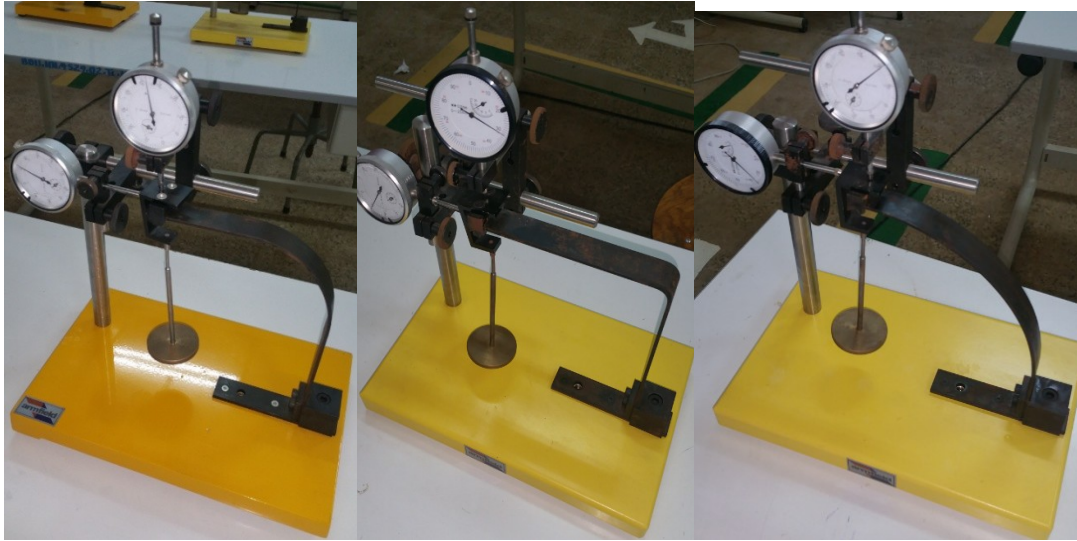


Figure 8: curved beam apparatus with different beam profile

4.3.1. Apparatus

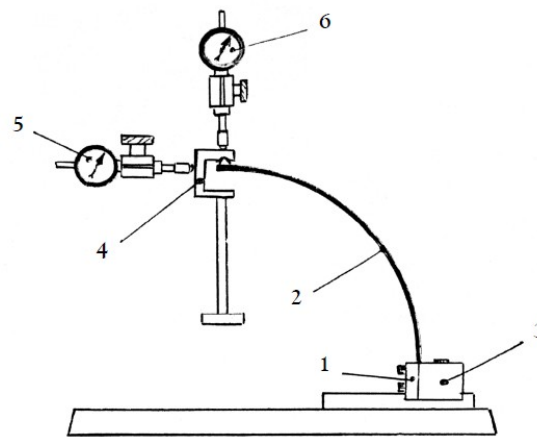


Figure 9: Curved beam deflection apparatus

4.3.2 The Specimens

Material Steel 25.4mm × 3.2mm

Specimen 1 a = 75mm R = 75mm b = 75mm

Specimen 2 a = 0mm R = 150mm b = 0mm,

Specimen 3 a = 0mm R = 75mm b = 75mm,

Specimen 4 a = 150mm R = 0mm b = 150mm

4.3.3 Assembly

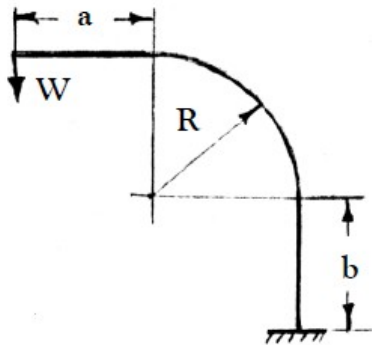
Instructions

- ❖ Unpack all components and layout on the work table.
- ❖ Place base Plate 5 on a level surface.
- ❖ Fit Vertical Column 6 to Base Plate 5 and lock in position with the socket head screw provided.
- ❖ Fit dial gauge carrier assembly 7 to the vertical Column 6.
- ❖ Select specimen 2 to be tested.
- ❖ Follow instructions for Procedure on page 3 of the Instruction Manual

2.4 Procedure

- ❖ Release clamp 1 and place specimen 2 in position.
- ❖ Clamp specimen.
- ❖ Release block 3 and Reposition if necessary to suit selected specimens.
- ❖ Lock in the required position.
- ❖ Place special load hanger 4 on the specimen.
- ❖ Position dial indicator supports so that the indicators 5 & 6 make contact with the load hanger 4 as indicated.
- ❖ Set indicators to zero.
- ❖ Load specimen by placing a weight on hanger 4 and note the dial indicator readings.
- ❖ Increase the load by equal increments and note the indicator readings for each load.

4.3.4 Sample example (Specimen 1)



$$a = 75\text{mm}$$

$$R = 75\text{mm}$$

$$b = 75\text{mm}$$

- 1 Plot the values in the diagram.
- 2 Compare the Theoretical and Experimental values.

Load (Kg)	Deflection Δ_W	Deflection Δ_P (mm)
0.2	6.8	3.2
0.4	12.8	6.4
0.6	18.8	9.6
0.8	25.5	12.8
1.0	31.8	15.7

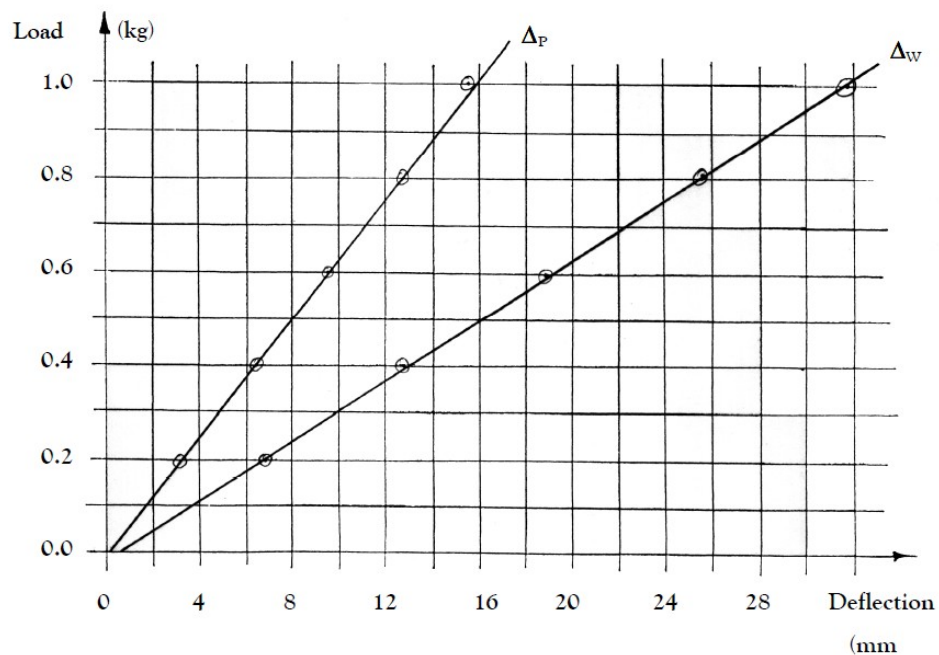
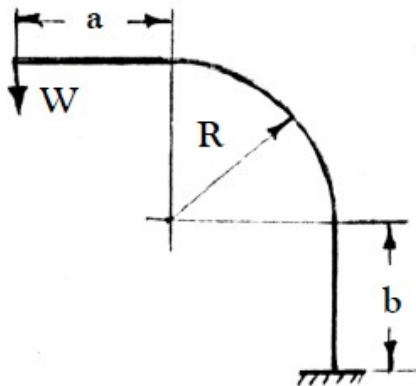


Figure 10: Figure 11: load Vs deflection graph (Δ_W & Δ_P)

5. Laboratory experiment

5.1. Exercise 1 (Specimen 1)



$$a = 75\text{mm}$$

$$R = 75\text{mm}$$

$$b = 75\text{mm}$$

- 1 Plot the values in the diagram.
- 2 Compare the Theoretical and Experimental values.

Table 1: input and output results of load and deflection

No	Load (Kg)	Deflection ΔW (mm)	Deflection ΔP (mm)

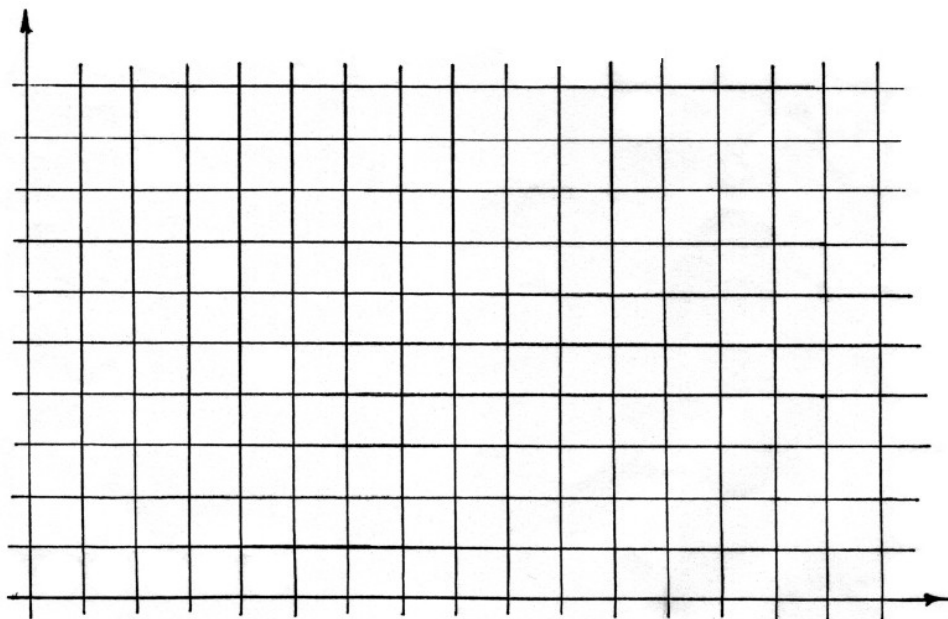
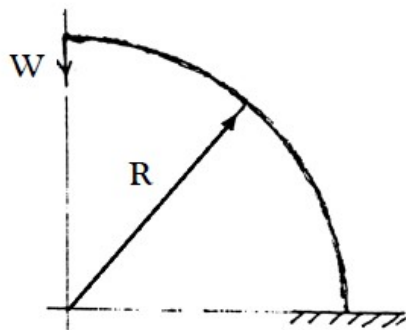


Figure 12: load Vs deflection graph (ΔW & ΔP)

5.2. Exercise 2 (Specimen 2)

Table 2: input and output results of load and deflection

No	Load (Kg)	Deflection ΔW (mm)	Deflection ΔP (mm)



$$\begin{aligned} a &= 0\text{mm} \\ R &= 150\text{mm} \\ b &= 0\text{mm} \end{aligned}$$

- 1 Plot the values in the diagram.
- 2 Compare the Theoretical and Experimental values.

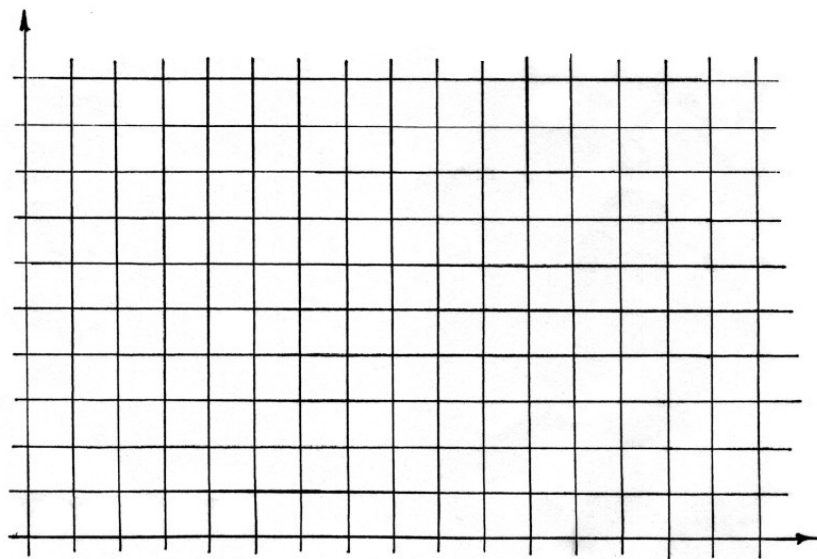
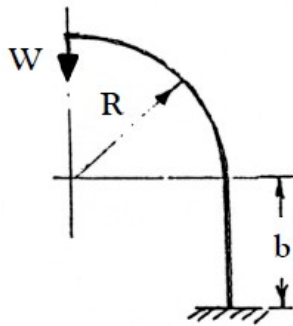


Figure 13: Figure 14: load Vs deflection graph (ΔW & ΔP)

5.3. Exercise 3 (Specimen 3)



$$\begin{aligned} a &= 0\text{mm} \\ R &= 75\text{mm} \\ b &= 75\text{mm} \end{aligned}$$

- 1 Plot the values in the diagram.
- 2 Compare the Theoretical and Experimental values.

Table 3: input and output results of load and deflection

No	Load (Kg)	Deflection $\Delta W(\text{mm})$	Deflection $\Delta P(\text{mm})$

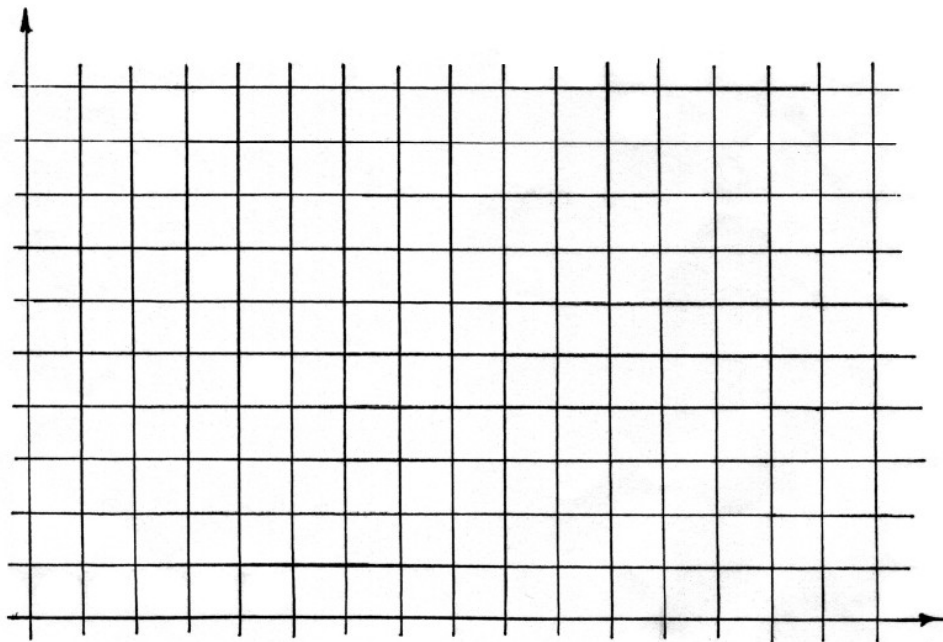
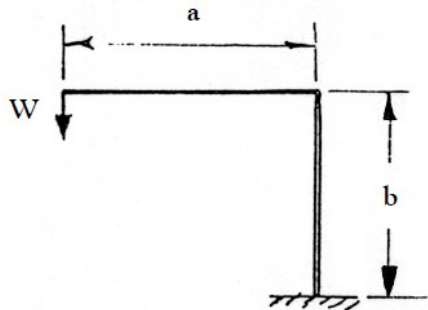


Figure 15: Figure 16: load Vs deflection graph (ΔW & ΔP)

5.4. Experiments 4(Specimen 4)

Exercise 4 (Specimen 4)



$$a = 150\text{mm}$$

$$R = 0\text{mm}$$

$$b = 150\text{mm}$$

- 1 Plot the values in the diagram.
- 2 Compare the Theoretical and Experimental values.

Table 4: input and output results of load and deflection

No	Load (Kg)	Deflection $\Delta W(\text{mm})$	Deflection $\Delta P(\text{mm})$

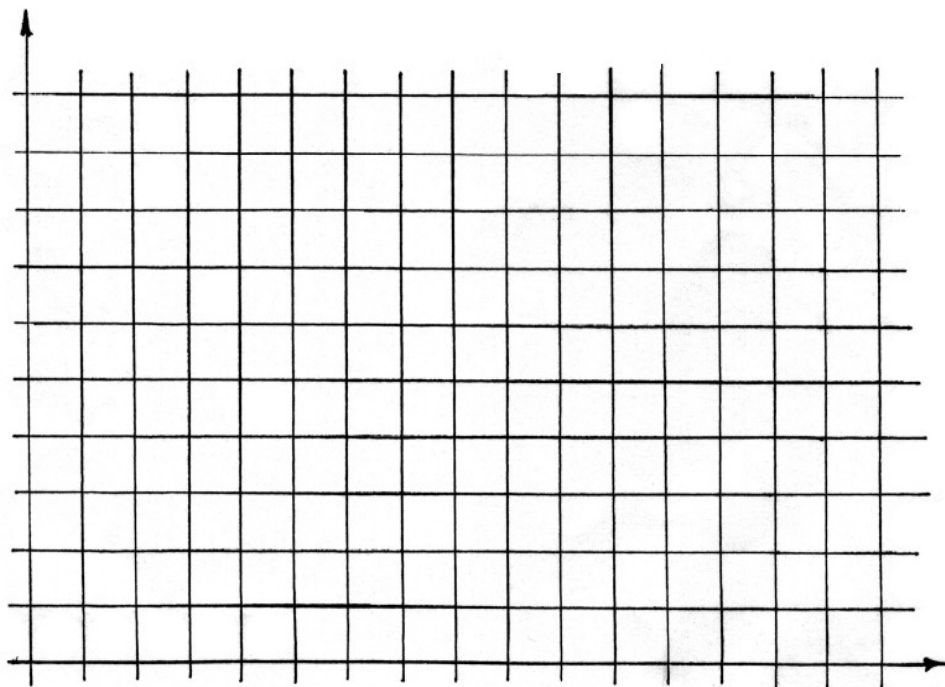


Figure 17: Figure 18: load Vs deflection graph (ΔW & ΔP)

6. Appendix

Unsymmetrical deflection of beam

Worksheet 1:

3.3.6 Practical exercise 1: Determination of the deflection of the beam

1. Record the results obtained:

Type of beam:

	0.5 kg		1.0 kg		1.5 kg	
ANGLE (°)	Δu (mm)	Δv (mm)	Δu (mm)	Δv (mm)	Δu (mm)	Δv (mm)
0						
15						
30						
45						
60						
75						
90						
105						
120						
135						
150						
165						
180						

*Worksheet 2:***3.4.4 Practical exercise 2: Determination of the shear center of a cantilever beam****2. Record the results obtained:**

Type of beam:

	0.5 kg		1.0 kg		1.5 kg	
ANGLE (°)	Δu (mm)	Δv (mm)	Δu (mm)	Δv (mm)	Δu (mm)	Δv (mm)
0						
15						
30						
45						
60						
75						
90						
105						
120						
135						
150						
165						
180						

Worksheet 3: Deflection of curved beam**5.1. Exercise 1, spacemen 1**

No	Load (Kg)	Deflection DW (mm)	Deflection DP (mm)

5.2. Exercise 2, spacemen 2

No	Load (Kg)	Deflection DW (mm)	Deflection DP (mm)

5.3. Exercise 3, spacemen 3

No	Load (Kg)	Deflection DW (mm)	Deflection DP (mm)

5.4. Exercise 4, spacemen 4

No	Load (Kg)	Deflection DW (mm)	Deflection DP (mm)

7. Reference

1. Edibon Engineering and technical teaching equipment manufacturing
2. Armfield engineering teaching and research Equipment
3. Strength of material books
4. Deferent lab Manual preparation website, MOS, MT lab manual preparation